

2 Wiring Diagrams

2.1 Magnet System Wiring

Illustration 6-5: Magnet System Wiring: Sleeve Connections

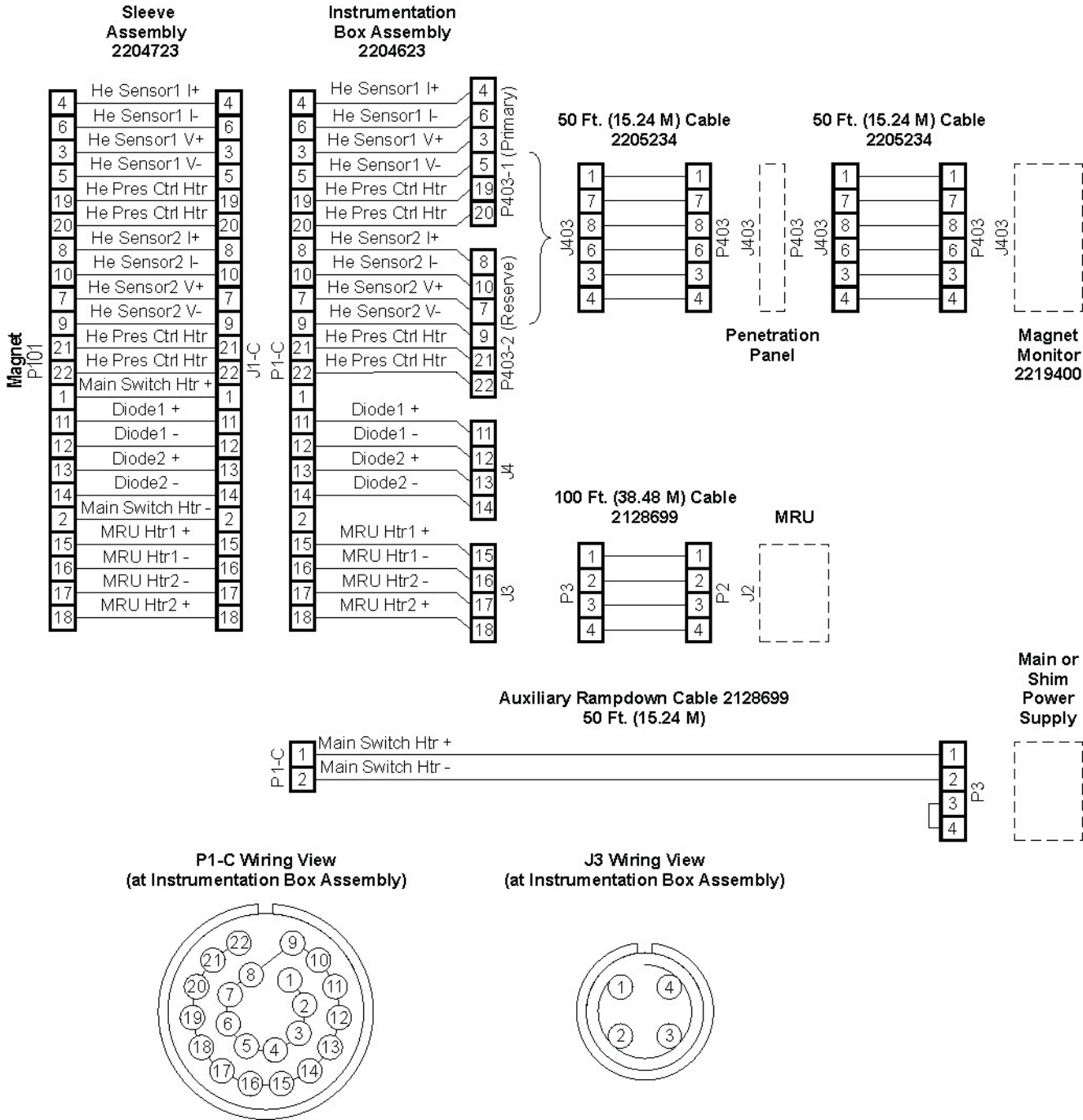


Illustration 6-6: Magnet System Wiring: Shim Lead Connections

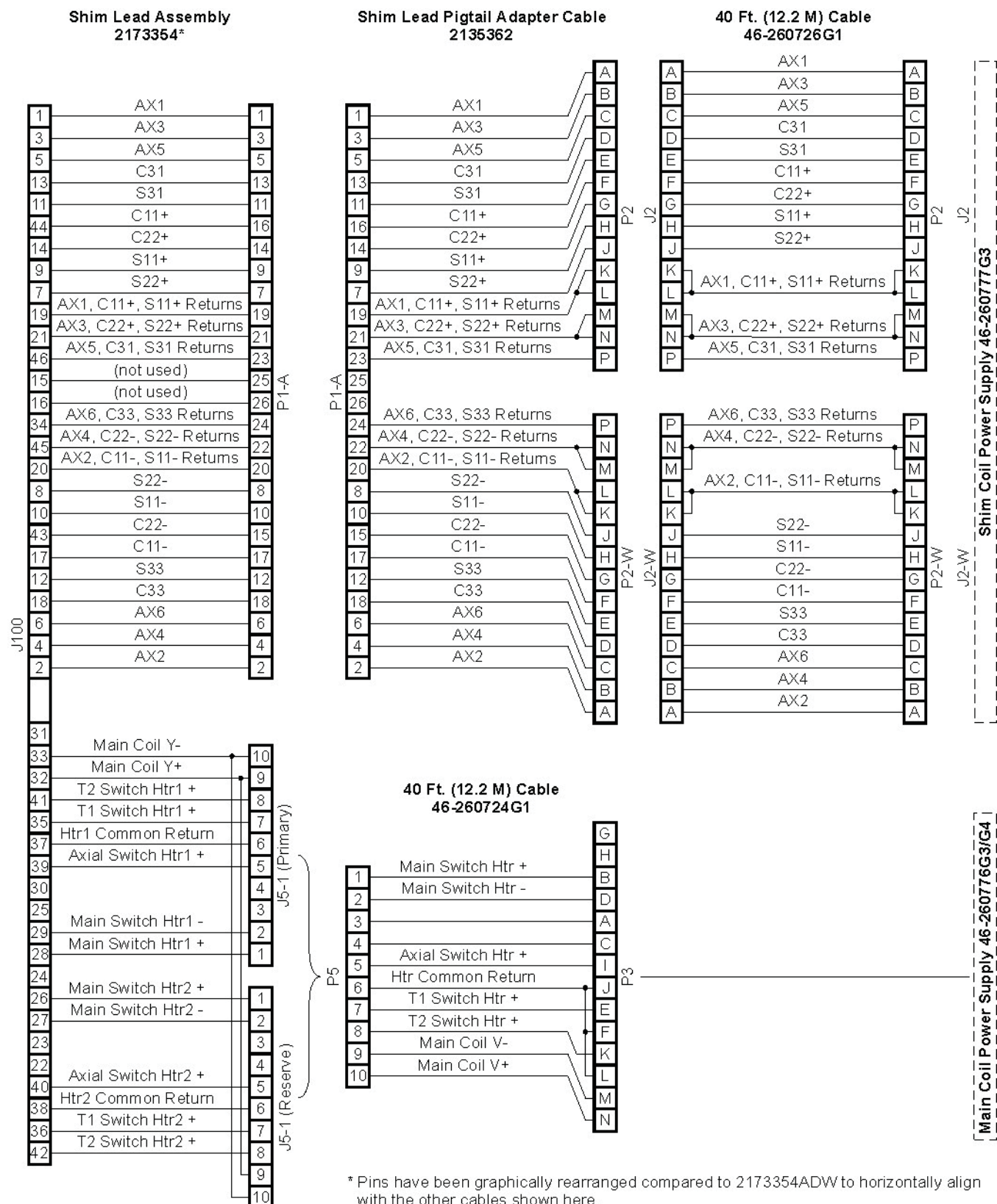


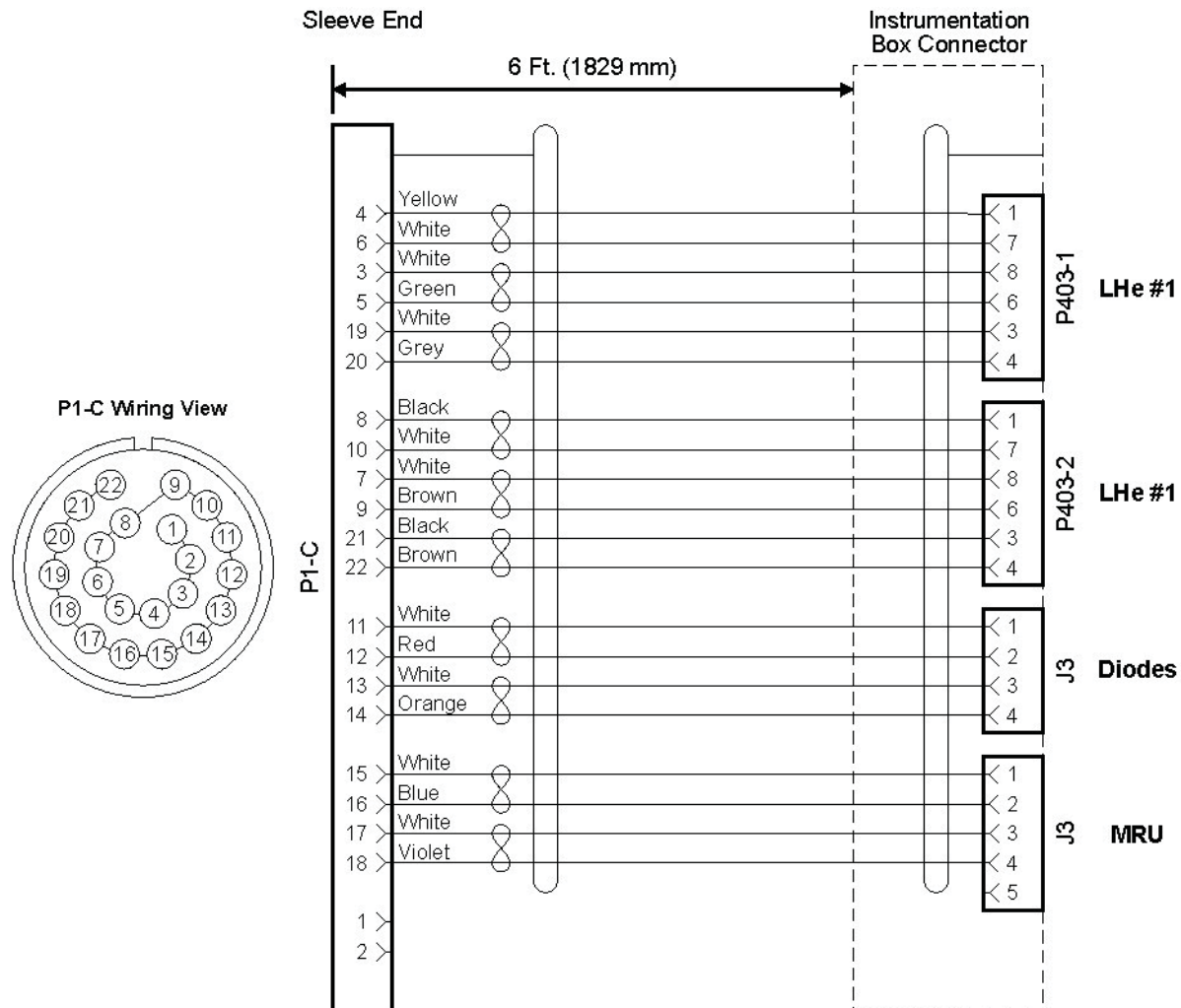
Illustration 6-7: Instrumentation Box Assembly 2204623 Wiring

Illustration 6-8: Shim Lead Pigtail Adapter Cable 2135362 Wiring

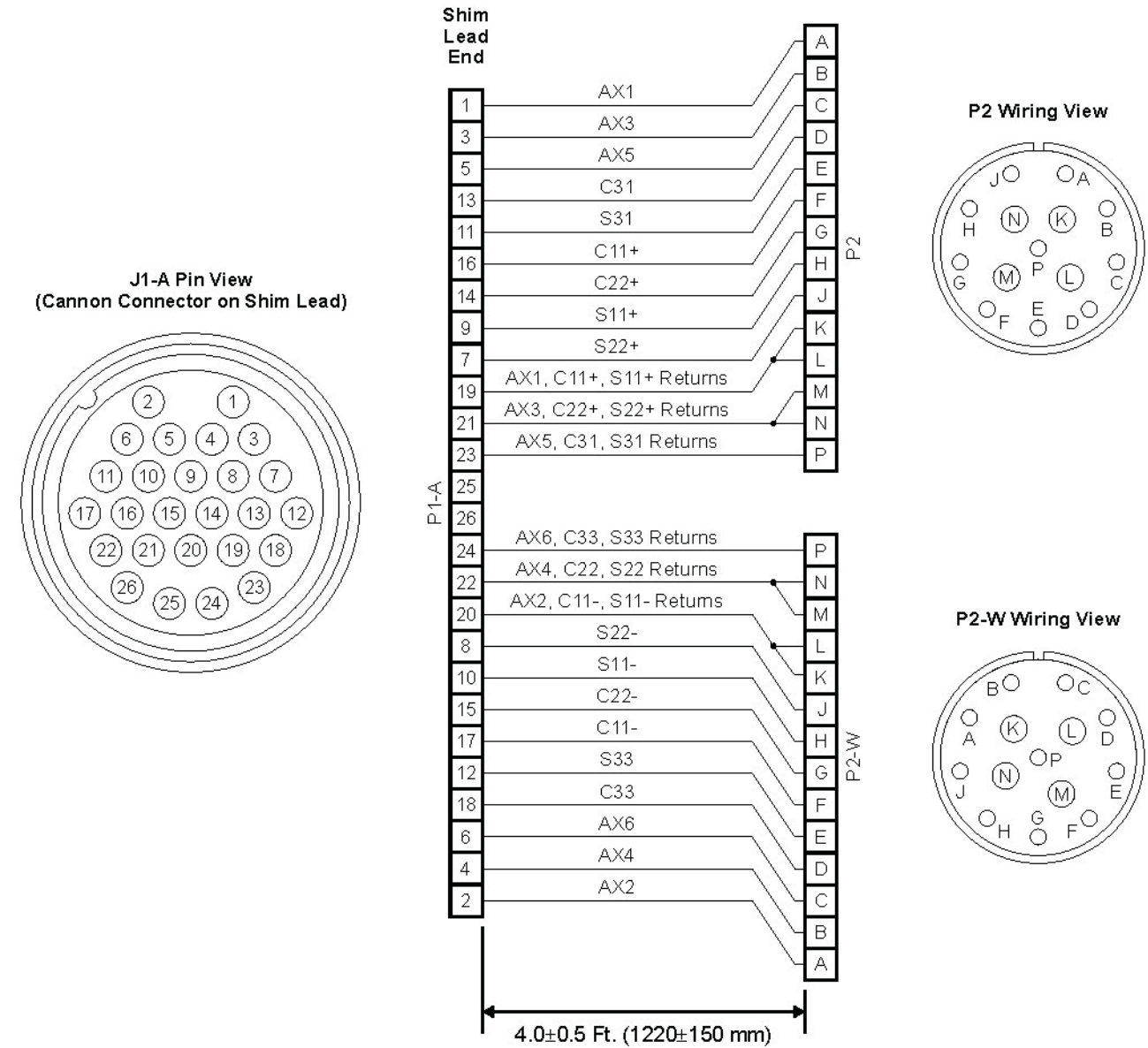


Illustration 6-9: Shim Coil Power Supply Cable 46-260726G1

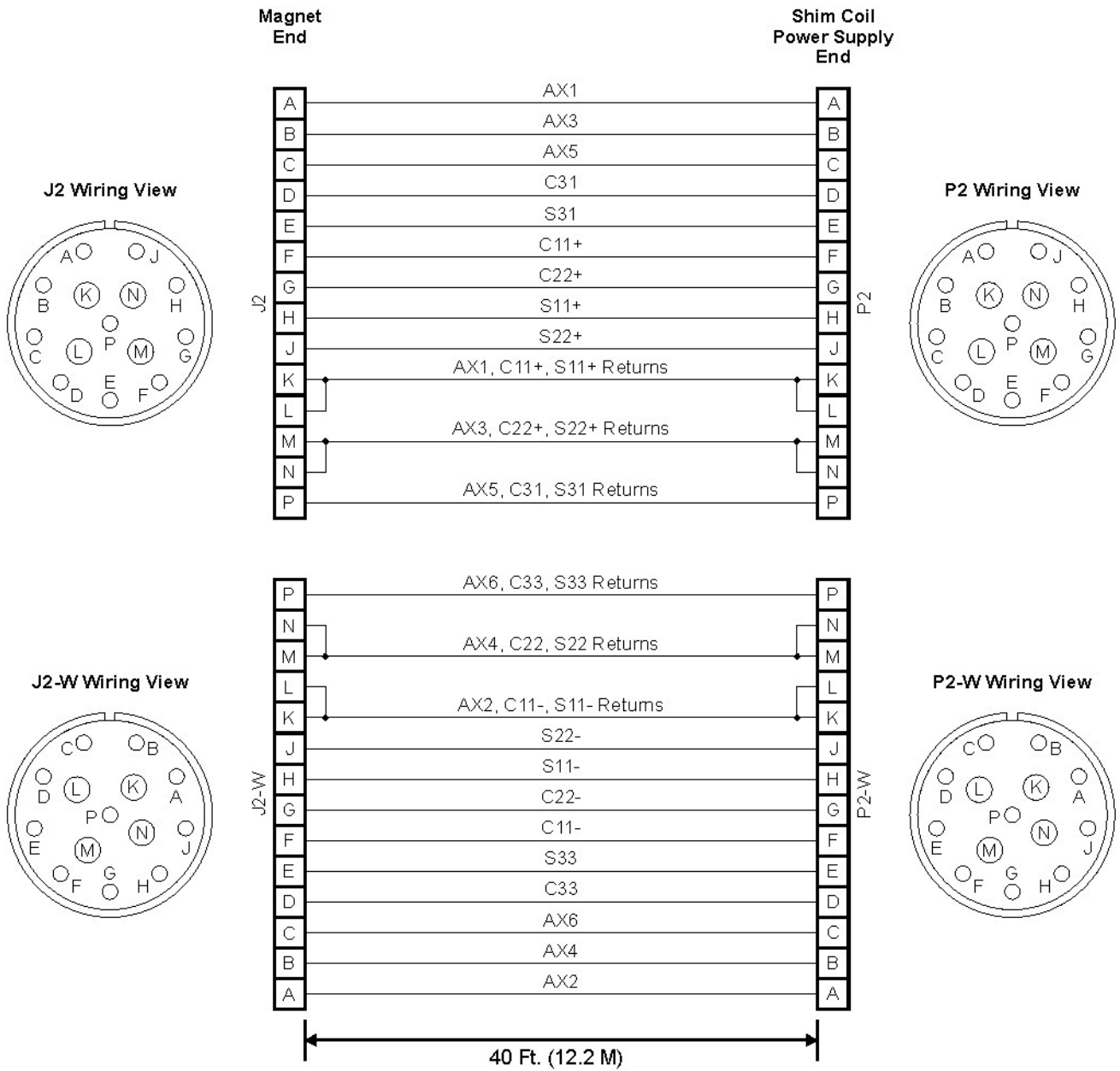


Illustration 6-10: Liquid Helium Monitor Circuit Wiring (Cables 2205234)

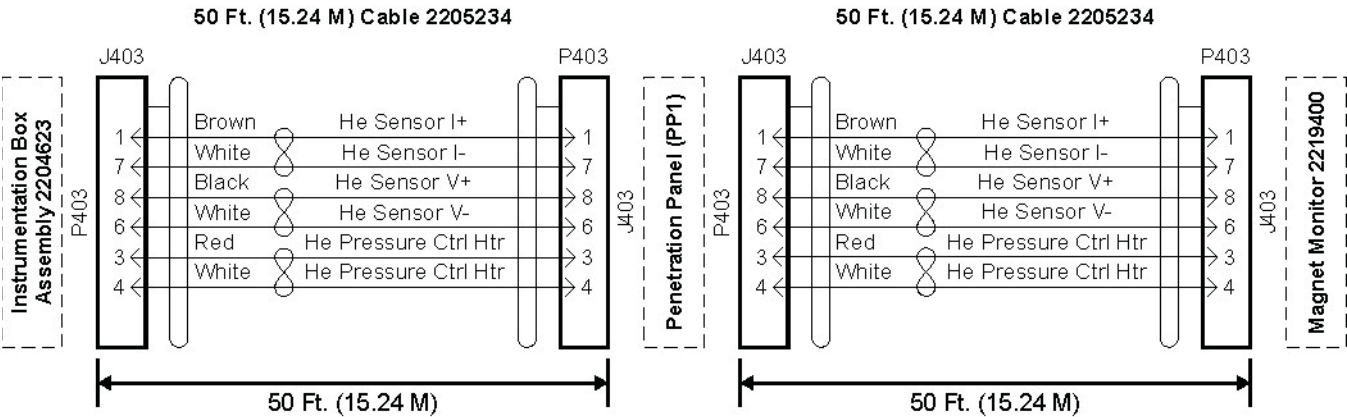
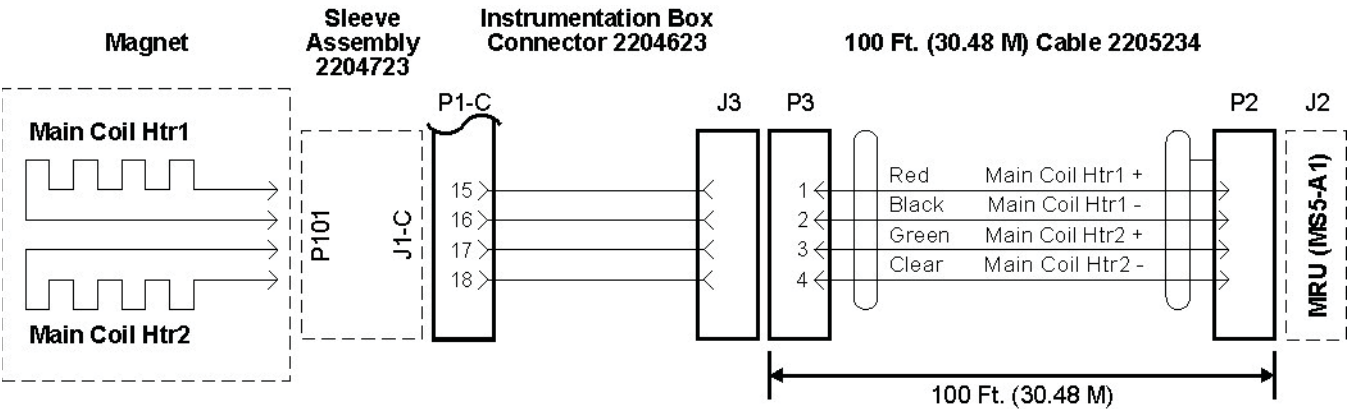


Illustration 6-11: Main Coil Heater Rundown Wiring



2.2 RuO Circuits and Pinouts

Illustration 6-12: RuO Circuit & Pinouts, Recondenser Vacuum Sleeve

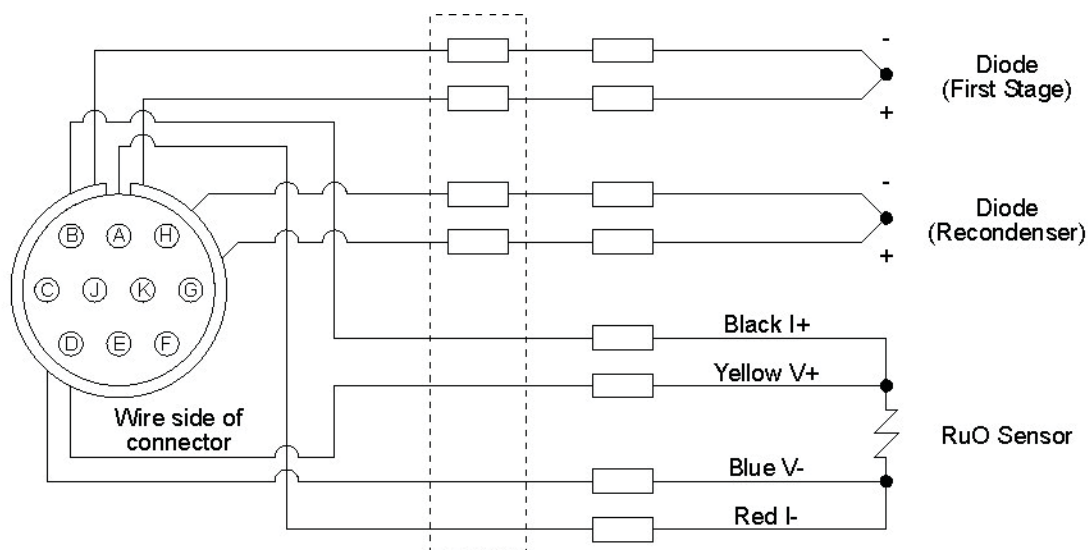
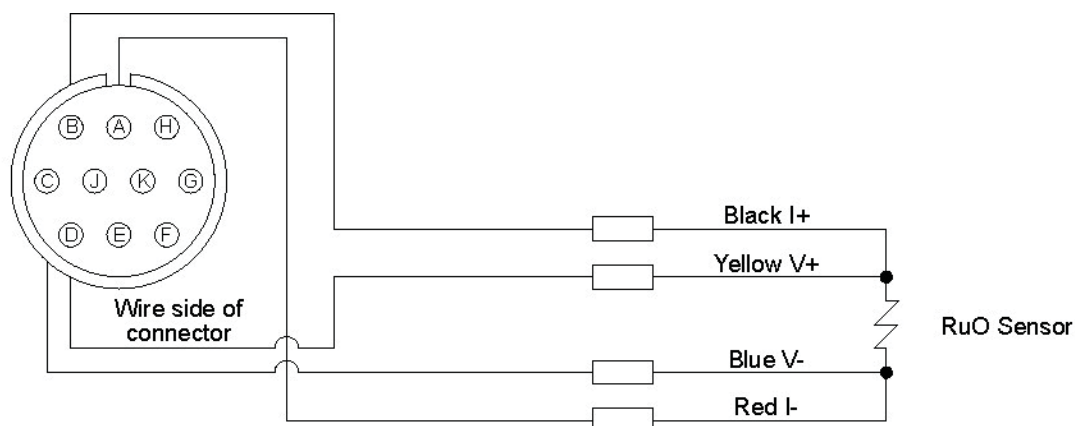


Illustration 6-13: RuO Circuit & Pinouts, Coldhead Transition Flange



2.3 Magnet Monitor Cables

Illustration 6-14: Magnet Monitor Cable A (Run 832): 2214062

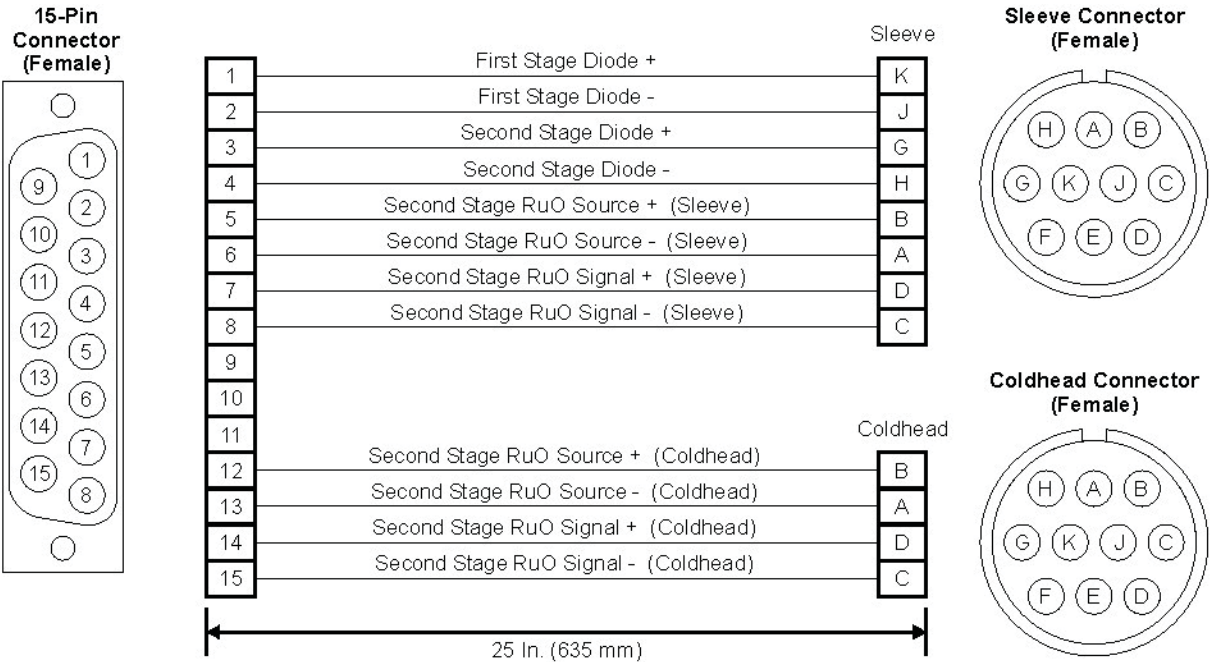


Illustration 6-15: Magnet Monitor Cables D (Run 828) and H (Run 824): 46-271609G29

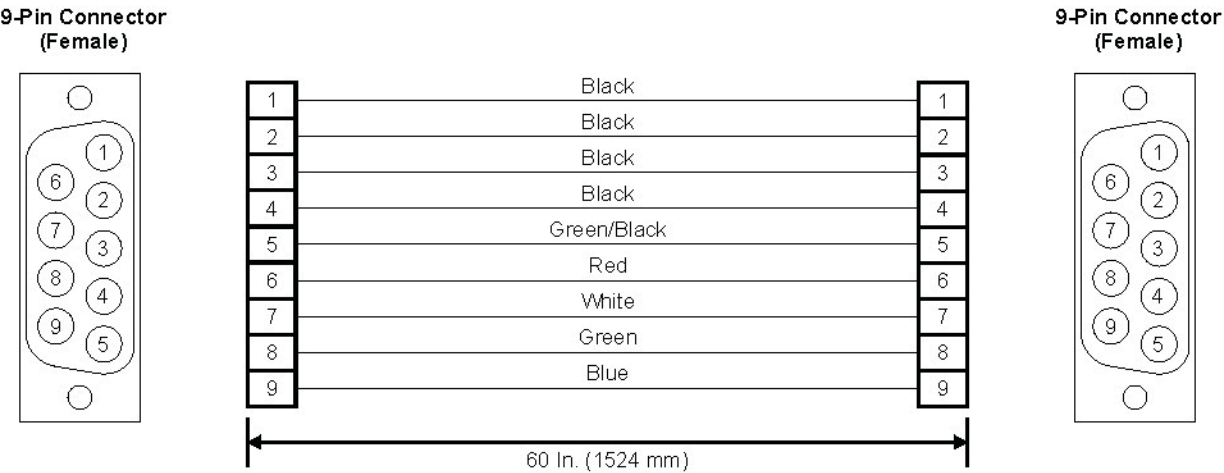


Illustration 6-16: Magnet Monitor Cable C (Run 829): 46-271601G29

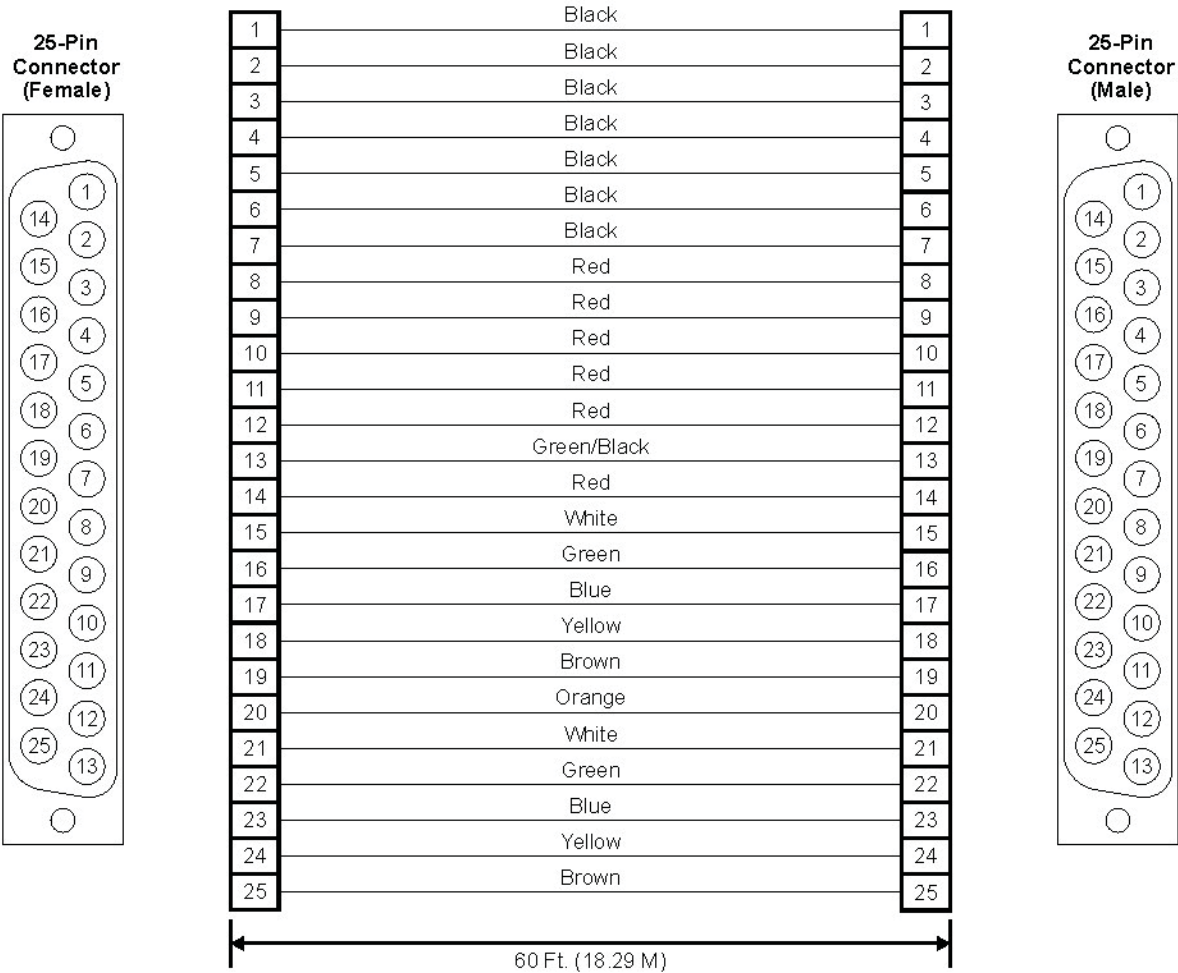


Illustration 6-17: Magnet Monitor Cable B (Run 830): 2222797

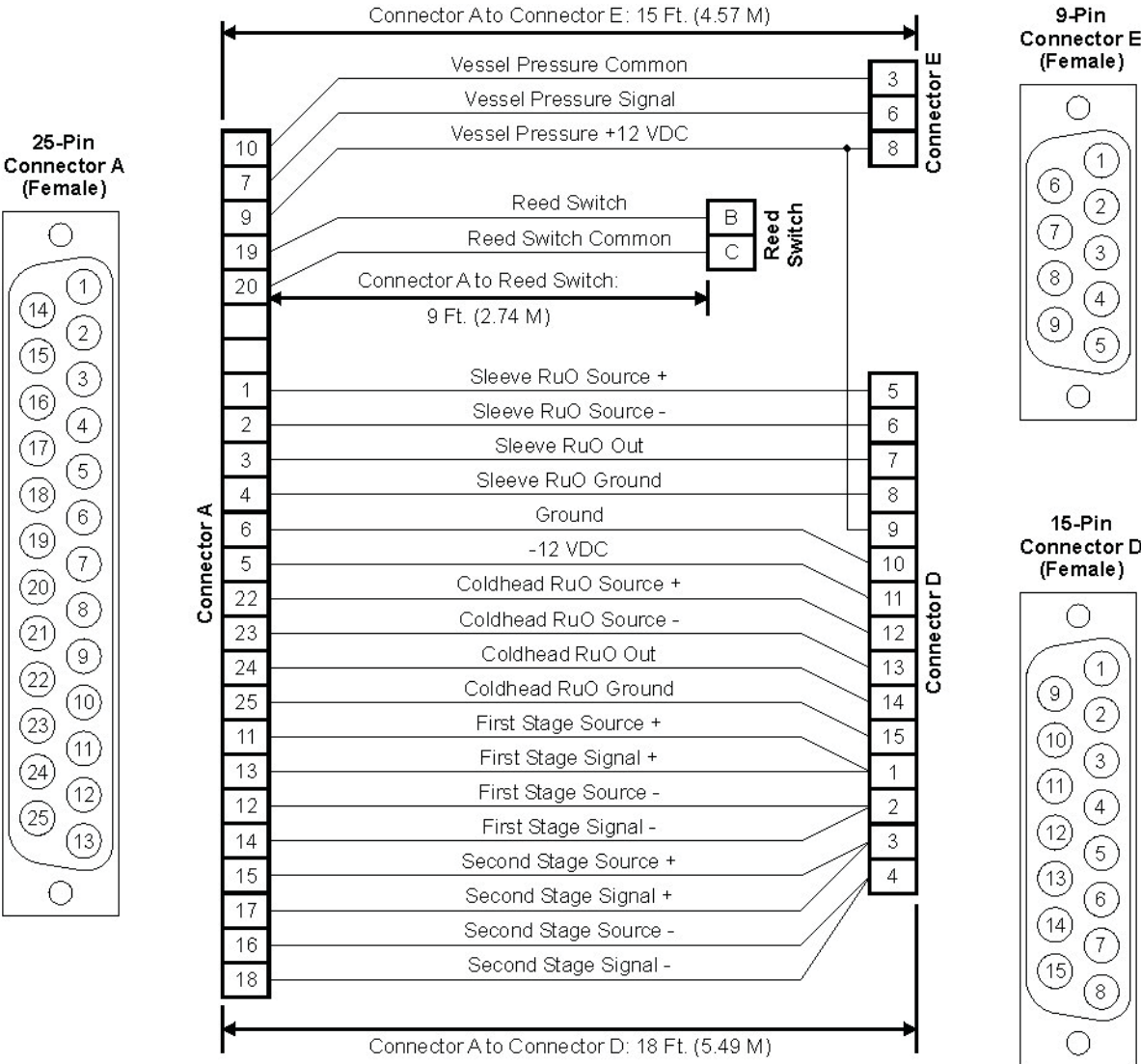


Illustration 6-18: Magnet Monitor Cable E (Run 823) and K (Run 826): 46-271603G29

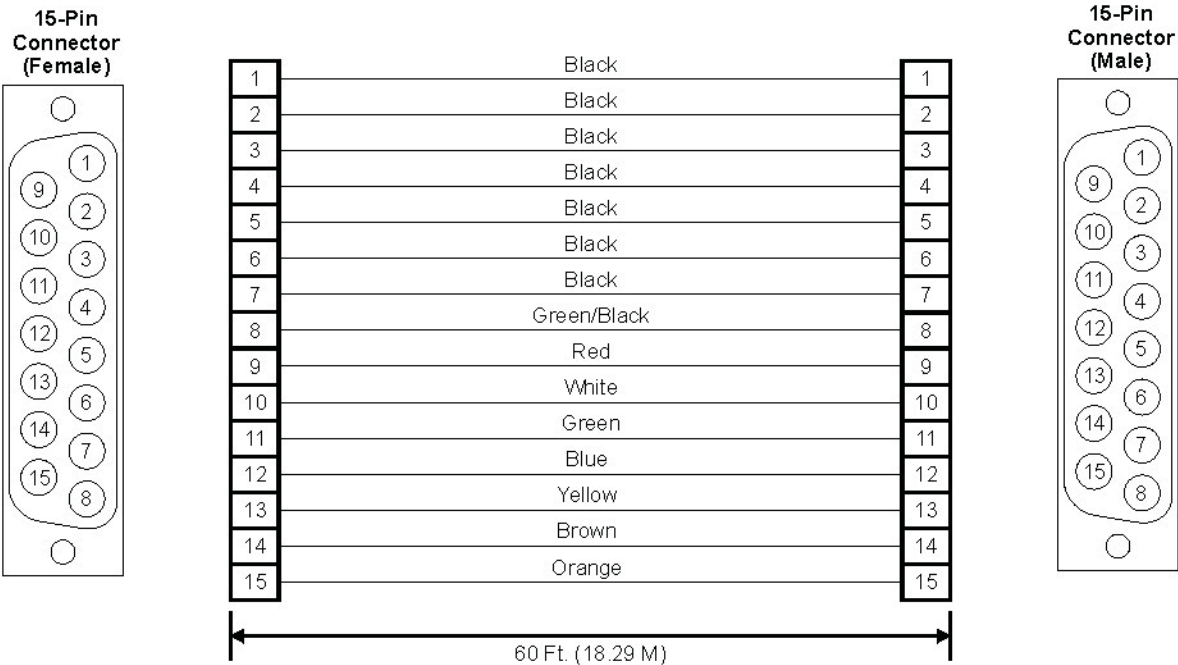


Illustration 6-19: Magnet Monitor Cable F: 2204485

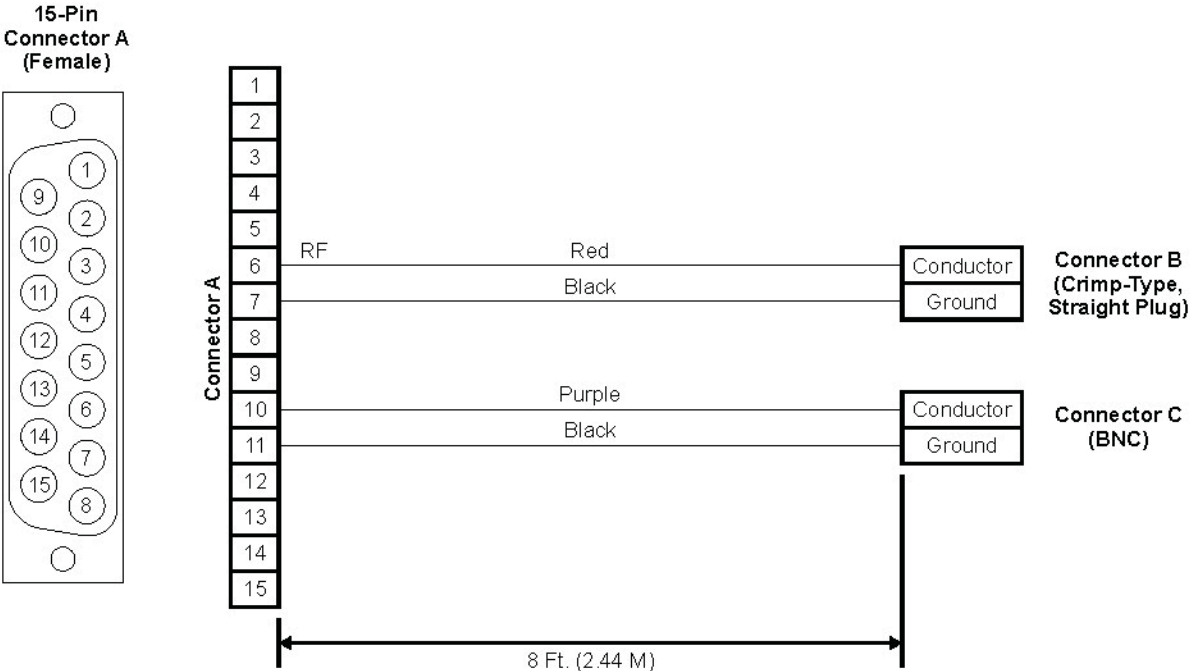


Illustration 6-20: Magnet Monitor Cable G (Run 831): 2204488-5

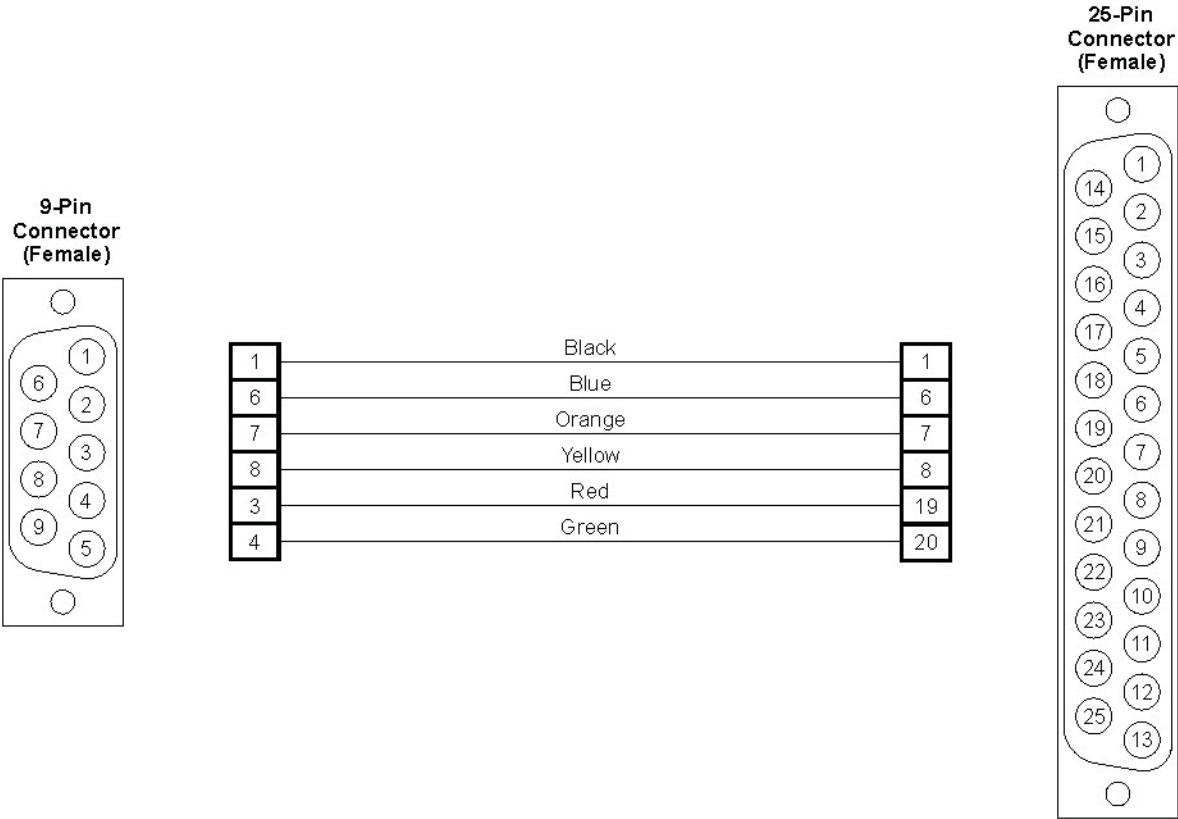


Illustration 6-21: Magnet Monitor Cable L (Run 827): 46-328578P1

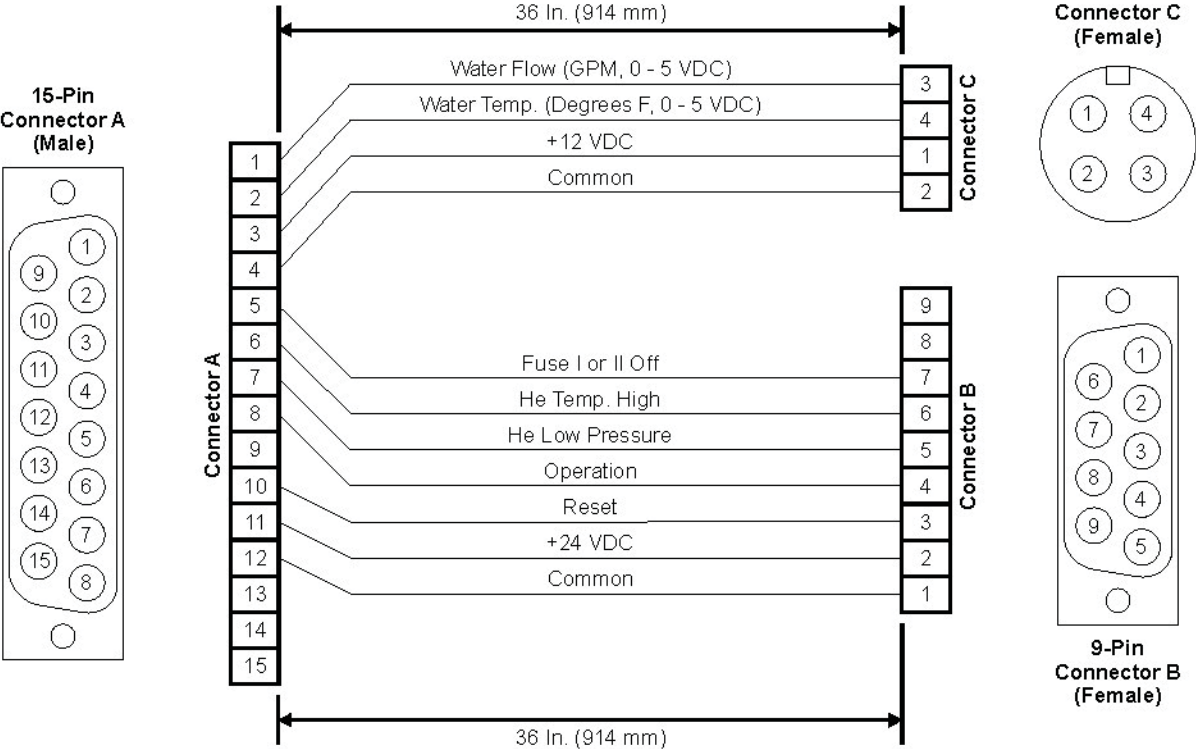


Illustration 6-22: Magnet Monitor Cable J (Run 825): 46-271600G29

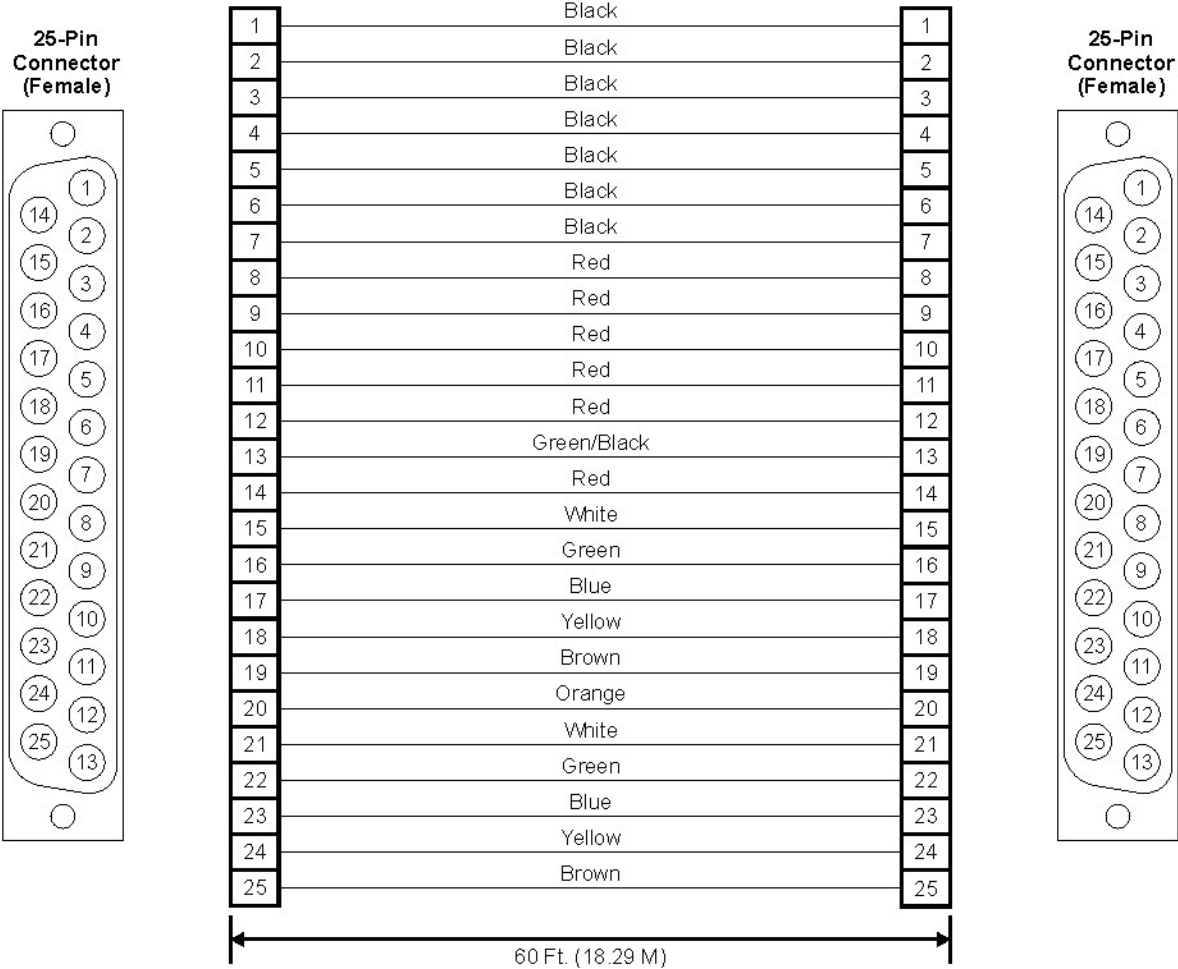
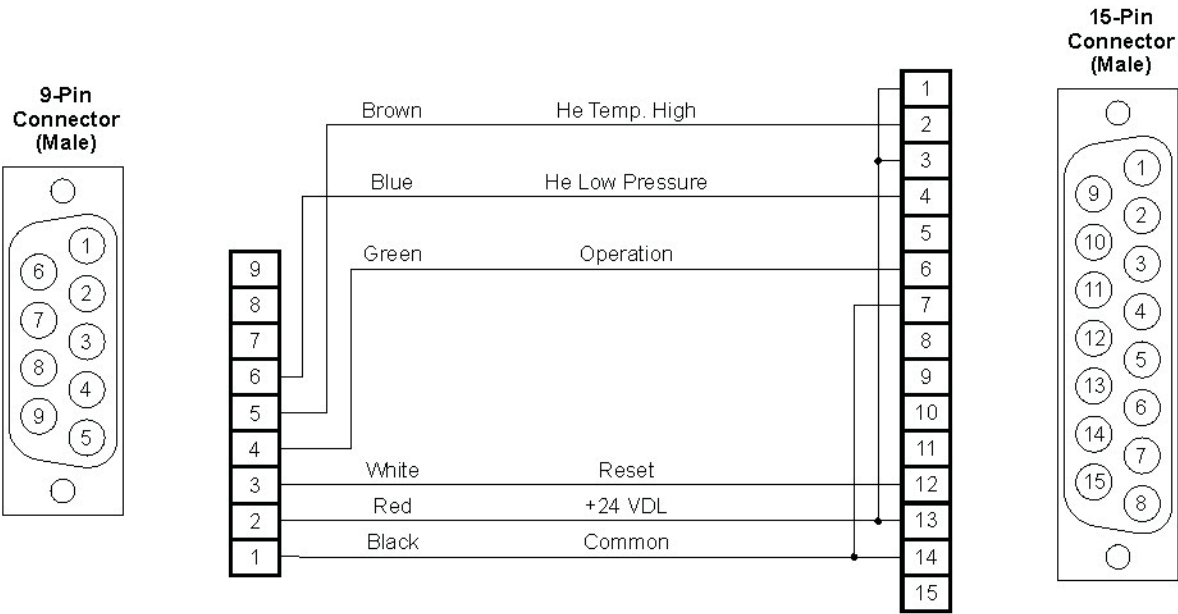


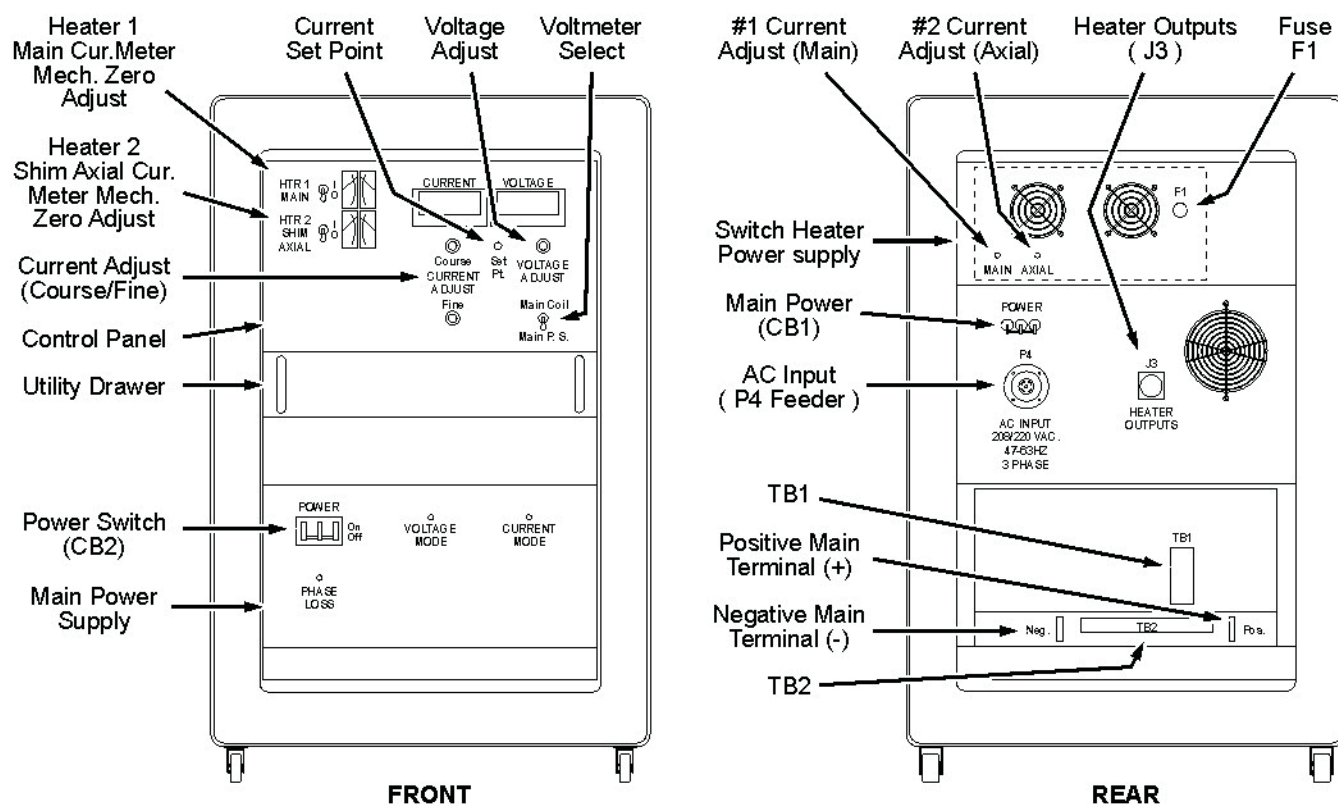
Illustration 6-23: Magnet Monitor Cable M (Run 833): 2225805-1



3 Power Supplies

3.1 750 Amp Main Coil Power Supply

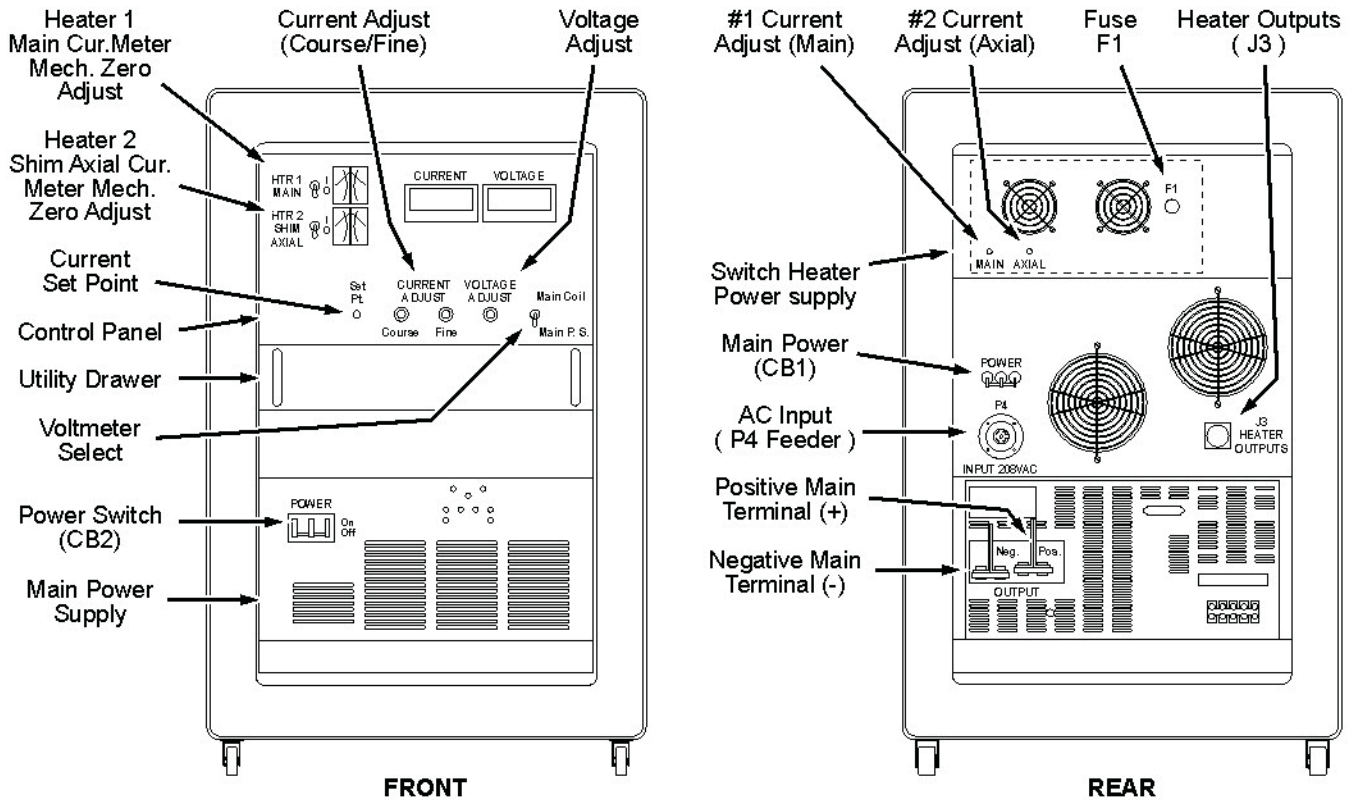
Illustration 6-24: 750 Amp Superconducting Main Coil Service Power Supply Cabinet Model TCR 7.5T750 (46-260776G3)



	Designator	Rating and Type	Connection
Input Power	P4	208/220 VAC, 50 - 60 Hz	—
	TB1	30 Amp, 602Y	AC Input, Main Power Supply
Output Connections	Main Terminal Positive (+)	750 Amp Bus Bar	Red (+) Main Lead Extension (MS3-A2)
	Main Terminal Negative (-)	750 Amp Bus Bar	Black (-) Main Lead Extension (MS3-A2)
	J3	1 Amp, MS3106A20-27P	Heater Wire Harness (MS3-A5) P3
Fuses and Circuit Breakers	CB1 Cabinet	3 Pole, 25 Amp, 250 VAC	—
	CB1 Supply	3 Pole, 25 Amp, 250 VAC	
	F1	4 Amp, 250 VAC, MDA	

3.2 1000 Amp Main Coil Power Supply

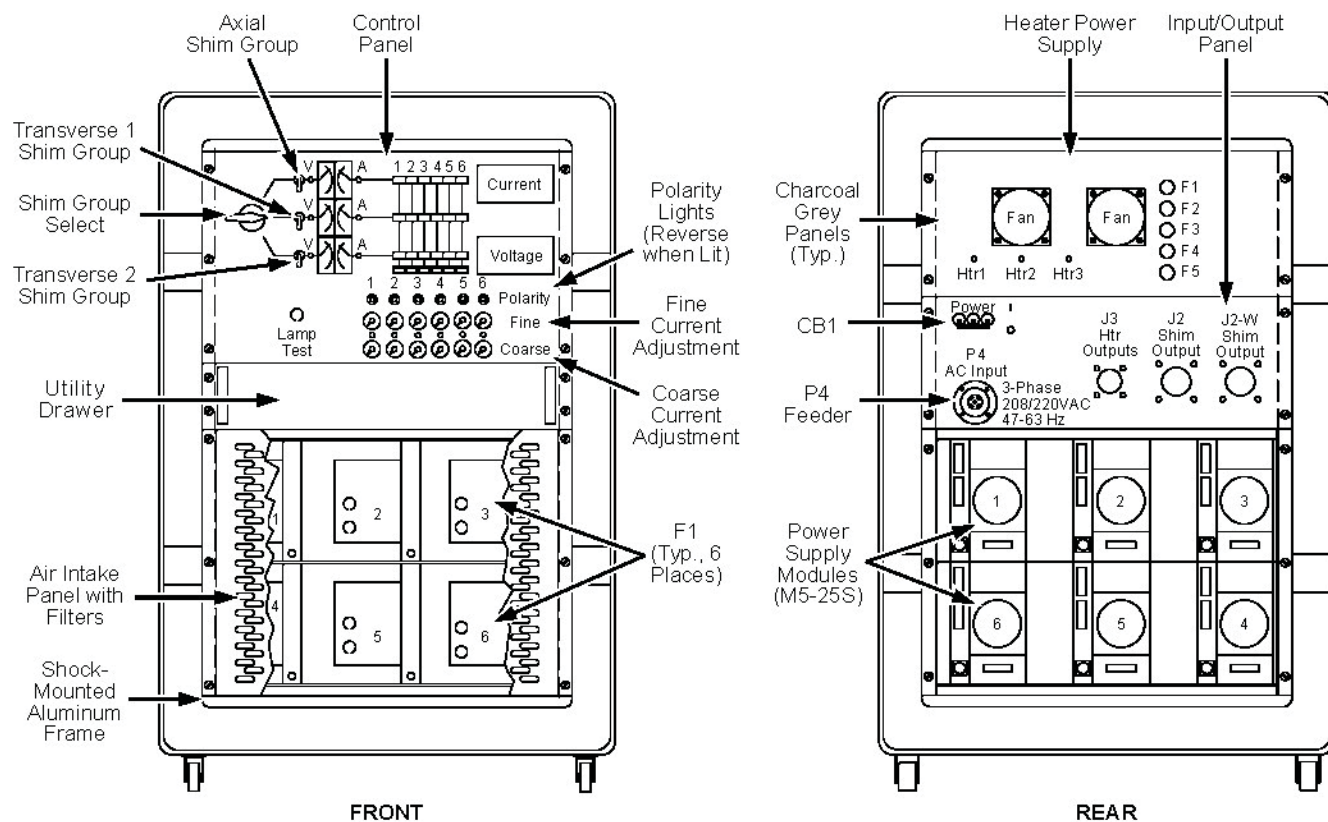
Illustration 6-25: 1000 Amp Superconducting Main Coil Service Power Supply Cabinet Model ESS 7.5-1000-2-D-1236 (46-260776G4)



	Designator	Rating and Type	Connection
Input Power	P4	208/220 VAC, 50 - 60 Hz	—
	TB1	30 Amp, 602Y	AC Input, Main Power Supply
Output Connections	Main Terminal Positive (+)	1000 Amp Bus Bar	Red (+) Main Lead Extension (MS3-A2)
	Main Terminal Negative (-)	1000 Amp Bus Bar	Black (-) Main Lead Extension (MS3-A2)
	J3	1 Amp, MS3106A20-27P	Heater Wire Harness (MS3-A5) P3
Fuses and Circuit Breakers	CB1 Cabinet	3 Pole, 25 Amp, 250 VAC	—
	CB1 Supply	3 Pole, 25 Amp, 250 VAC	
	F1	1 Amp, 250 VAC, MDA	

3.3 Shim Coil Power Supply

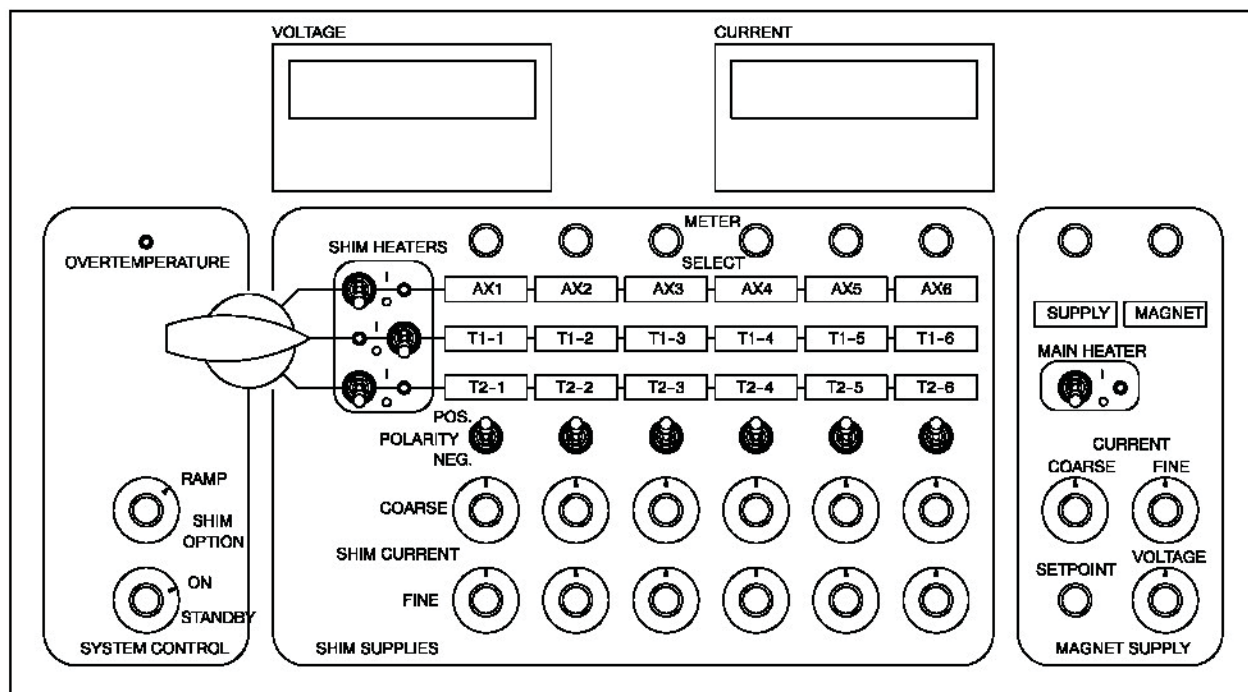
Illustration 6-26: Superconducting Shim Coil Service Power Supply Cabinet (46-260777G3)



	Designator	Rating and Type	Connection
Input Power	P4	208/220 VAC, 50 - 60 Hz	—
Output Connections	J2	25 Amp, MS3012A28-20S	S/C Shim Coil Wire Harness (MS3-A4) P1
	J2-W	25 Amp, MS3012A28-20S	S/C Shim Coil Wire Harness (MS3-A4) P2
	J3	1 Amp, MS3106A20-27P	Heater Wire Harness (MS3-A5) P3
Fuses and Circuit Breakers	CB1	3 Pole, 25 Amp, 250 VAC	—
	F1 (Typ., 6 Places)	6.25 Amp, 250 Volt, MDA	
	F1, F5	4 Amp, 250 Volt, MDA	
	F2, F3, F4	1 Amp, 250 Volt, MDA	

3.4 Phase III-A Power Supply

Illustration 6-27: Magnet/Shim Phase III-A Power Supply Control Panel



3.5 Power Supply Connector Pinouts

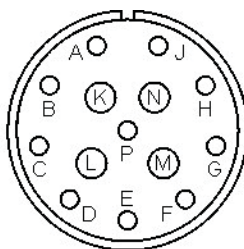
Illustration 6-28: P4 — Shim and Main Coil Power Supply AC Inputs



Pin	Signal
G	Ground
W	Neutral
X	Phase A
Y	Phase B
Z	Phase C

4 Pole, 5 Wire
Twist-Lock Flange Inlet
30A 120/208V 3 Phase Y
NEMA L21-30R

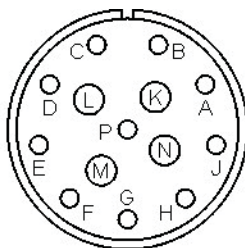
Illustration 6-29: J2 — Shim Coil Power Supply Output (MSA3102A28-20S)



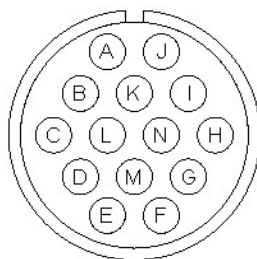
Pin	Signal
A	AX1
B	AX3
C	AX5
D	T1-1
E	T2-1

Pin	Signal
F	T1-2
G	T1-3
H	T2-2
J	T2-3
K/L*	AX1, T1-2, T2-2 Returns
M/N*	AX3, T1-3, T2-3 Returns
P	AX5, T1-1, T2-1 Returns
* Pins paralleled.	

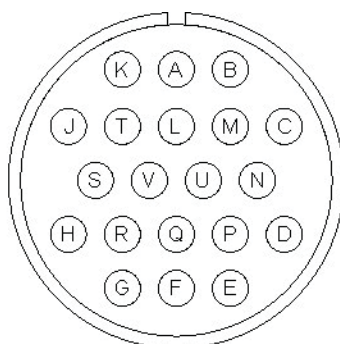
Illustration 6-30: J2-W — Shim Coil Power Supply Output (MSA3102A28-20S)



Pin	Signal
A	AX2
B	AX4
C	AX6
D	T1-6
E	T2-6
F	T1-4
G	T1-5
H	T2-4
J	T2-5
K/L*	AX2, T1-4, T2-4 Returns
M/N*	AX4, T1-5, T2-5 Returns
P	AX6, T1-6, T2-6 Returns
* Pins paralleled.	

Illustration 6-31: J3 — Switch Heater Output (MSA3102A20-27S)

Phase III Power Supply		Shim Coil Service Power Supply		Main Coil Service Power Supply	
Pin	Signal	Pin	Signal	Pin	Signal
A	—	A	—	A	—
B	Main Switch Htr +	B	—	B	Main Switch Htr +
C	—	C	—	C	—
D	Main Switch Htr -	D	—	D	Main Switch Htr -
E	Transverse 1 +	E	Transverse 1 +	E	—
F	Transverse 1 -	F	Transverse 1 -	F	—
G	—	G	—	G	—
H	—	H	—	H	—
I	Axial Switch Htr +	I	Axial Switch Htr +	I	Axial Switch Htr +
J	Axial Switch Htr -	J	Axial Switch Htr -	J	Axial Switch Htr -
K	Transverse 2 +	K	Transverse 2 +	K	—
L	Transverse 2 -	L	Transverse 2 -	L	—
M	Main Coil V+	M	—	M	Main Coil V+
N	Main Coil V-	N	—	N	Main Coil V-

Illustration 6-32: Optional Shim Coil Power Supply Output (MSA3102A20-27S)

Pin	Signal	Pin	Signal
A	Channel A = ZY	L	Channel D = X2-Y2
B	ZY Return	M	X2-Y2 Return
C	—	N	—
D	—	P	—
E	Channel B = XY	Q	Channel F = Z3
F	XY Return	R	Z3 Return
G	Channel C = ZX	S	Channel E = Z2
H	ZX Return	T	Z2 Return
J	—	U	—
K	—	V	—